

INSTRUCTIONS:

1. All questions are compulsory.
2. Select the most appropriate option for multiple-choice questions.

SECTION A: OBJECTIVE QUESTIONS

1. What is the primary function of the Transformer architecture's multi-head self-attention mechanism?  
(A) To perform sequence recurrent memory retention across timesteps  
(B) To attend to information from different representation subspaces at different positions simultaneously  
(C) To reduce the dimensional size of embedding vectors  
(D) To eliminate the need for feed-forward neural layers
2. Which data structure provides average  $O(1)$  time complexity for insertion, deletion, and search operations?  
(A) Balanced Binary Search Tree (AVL)  
(B) Hash Table with uniform distribution  
(C) B-Tree Index  
(D) Min-Heap Priority Queue
3. In PostgreSQL, what is the key difference between an optimistic concurrency control model and row-level advisory locks?  
(A) Advisory locks prevent writes using MVCC tuple versioning exclusively  
(B) Advisory locks provide explicit application-level locking without locking actual table rows  
(C) Row-level advisory locks are automatically released upon any SELECT statement  
(D) Optimistic concurrency requires serial execution of all transactions
4. Which network protocol operates at Layer 4 of the OSI model and provides connectionless, low-latency transmission?  
(A) Transmission Control Protocol (TCP)  
(B) User Datagram Protocol (UDP)  
(C) Hypertext Transfer Protocol (HTTP)  
(D) Internet Protocol (IP)
5. In distributed computing, what property ensures that all cluster nodes see the same data at the same time?  
(A) Availability  
(B) Consistency  
(C) Partition Tolerance  
(D) Eventual Convergence

ANSWER KEY:

1. B
2. B
3. B
4. B
5. B